

Unit 8, Lesson 8: Solutions to Systems of Equations

Benchmarks

MA.8.AR.4.1 Given a system of two linear equations and specified set of possible solutions, determine which ordered pairs satisfy the system of linear equations.

MA.8.AR.4.2 Given a system of two linear equations represented graphically on the same coordinate plane, determine whether there is one solution, no solution, or infinitely many solutions.

Learning Targets

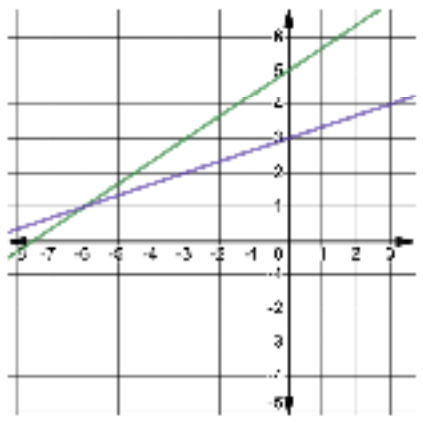
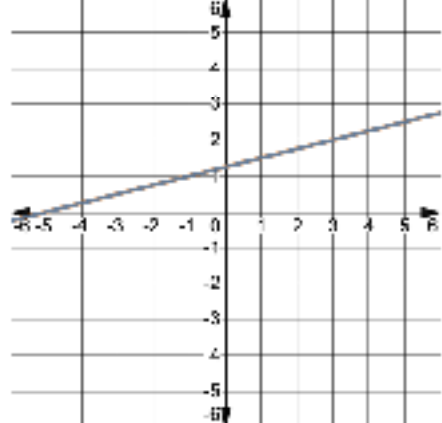
- ☐ I can determine algebraically if an ordered pair satisfies the given system of linear equations.
- ☐ I can verify graphically when a system of linear equations has one solution, no solution, or infinitely many solutions.

8.8.1 Warm-Up: Bell Work

Three systems of equations are shown.

System A	System B	System C
$\begin{cases} y = \frac{1}{4}x + \frac{5}{4} \\ y = 1.25 + 0.25x \end{cases}$	$\begin{cases} y = \frac{2}{3}x + 5 \\ y = \frac{1}{3}x + 3 \end{cases}$	$\begin{cases} y = -\frac{1}{8}x + 1 \\ y = -\frac{1}{6}x - 1 \end{cases}$

1. The graphs of each system are shown. Match each system of linear equations with its graphical representation.

Graph			
System			

8.8.2 Guided Instruction: Solutions to Systems of Equations

1. Consider the system of equations $\begin{cases} y = \frac{1}{2}x + 5 \\ y = -\frac{1}{4}x + 8 \end{cases}$.
- a. Determine algebraically, using substitution, if the ordered pair (2, 7.5) satisfies the system of linear equations.

Does (2, 7.5) satisfy the equation $y = \frac{1}{2}x + 5$?	Does (2, 7.5) satisfy the equation $y = -\frac{1}{4}x + 8$?
$\begin{aligned} y &= \frac{1}{2}x + 5 \\ \underline{\hspace{1cm}} &= \left(\frac{1}{2} \cdot \underline{\hspace{1cm}}\right) + 5 \\ \underline{\hspace{1cm}} &= \underline{\hspace{1cm}} + 5 \\ \underline{\hspace{1cm}} &= \underline{\hspace{1cm}} \end{aligned}$	$\begin{aligned} y &= -\frac{1}{4}x + 8 \\ \underline{\hspace{1cm}} &= \left(-\frac{1}{4} \cdot \underline{\hspace{1cm}}\right) + 8 \\ \underline{\hspace{1cm}} &= \underline{\hspace{1cm}} + 8 \\ \underline{\hspace{1cm}} &= \underline{\hspace{1cm}} \end{aligned}$

- b. Complete the statement.
- The ordered pair (2, 7.5)

☐ is

☐ is not

 a solution to the system.
- It

☐ does

☐ does not

 satisfy the equation $y = -\frac{1}{4}x + 8$, and
- it

☐ does

☐ does not

 satisfy the equation $y = \frac{1}{2}x + 5$. To be a solution to the system, the ordered pair

☐ only needs to satisfy one of the equations.

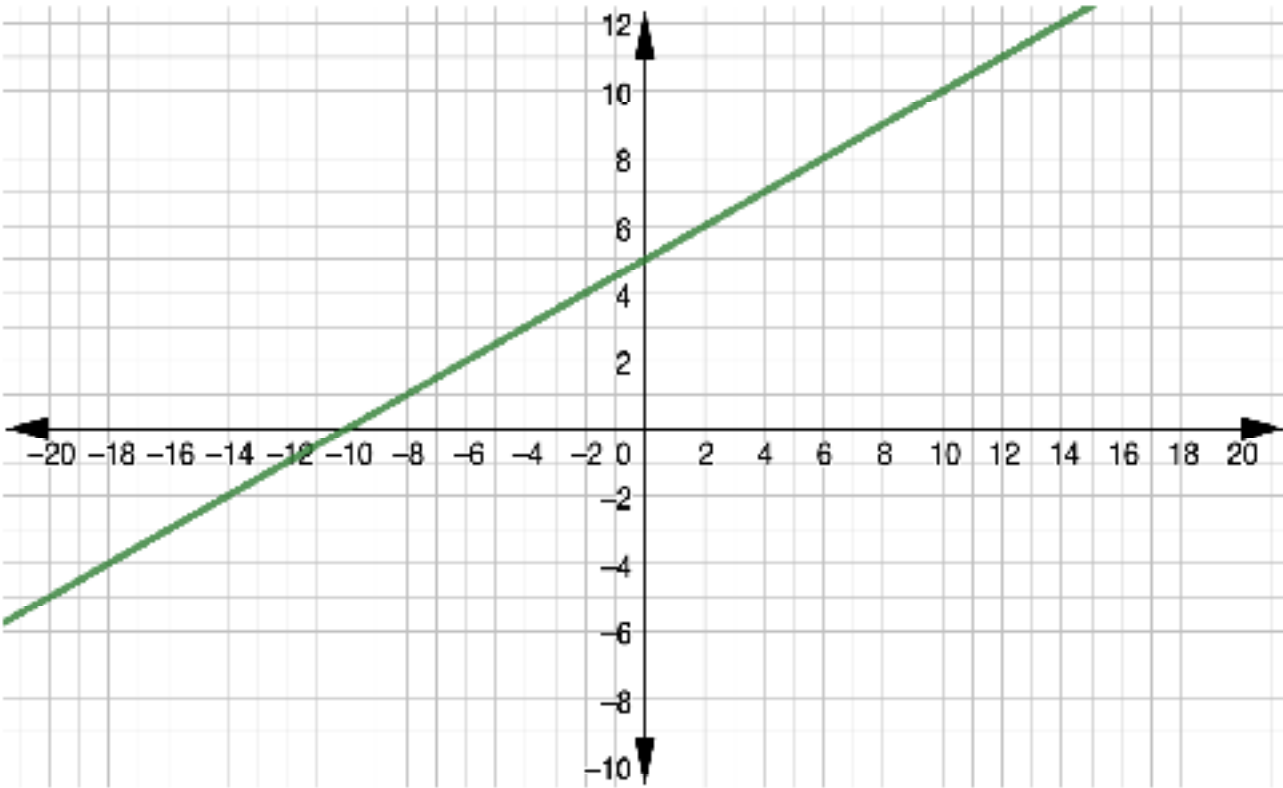
☐ must satisfy all equations in the system.

c. Complete the table to determine if each ordered pair satisfies both equations in the system.

Ordered Pair	$y = \frac{1}{2}x + 5$	$y = -\frac{1}{4}x + 8$	Are Both Equations True?
$(-2, 4)$			☐ Yes ☐ No
$(1, 8)$			☐ Yes ☐ No
$(4, 7)$			☐ Yes ☐ No

d. What is the solution to the system $\begin{cases} y = \frac{1}{2}x + 5 \\ y = -\frac{1}{4}x + 8 \end{cases}$?

2. The graph of $y = \frac{1}{2}x + 5$ is shown on the coordinate grid.



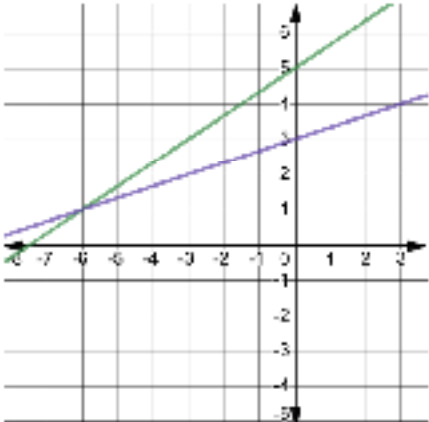
a. Graph $y = -\frac{1}{4}x + 8$ on the same coordinate grid.

b. Describe how the solution to the system $\begin{cases} y = \frac{1}{2}x + 5 \\ y = -\frac{1}{4}x + 8 \end{cases}$ determined in Part D of the previous question relates to the graph of the system.

c. How many solutions are there for this system?

Systems of equations in two variables can have one solution, no solution, or infinitely many solutions.

3. The systems of equations and their corresponding graphs from the Warm-Up are shown again in the table.
- a. Complete the description of the solution(s) for each system using the phrase one solution, no solution, or infinitely many solutions.

	System	Graph	Solution(s)
A	$\begin{cases} y = \frac{1}{4}x + \frac{5}{4} \\ y = 1.25 + 0.25x \end{cases}$		This system has _____ because the equations are equivalent. Every ordered pair on both lines is the same.
B	$\begin{cases} y = \frac{2}{3}x + 5 \\ y = \frac{1}{3}x + 3 \end{cases}$		This system has _____ because there is only one ordered pair that is on both lines. It is the point of intersection.
C	$\begin{cases} y = -\frac{1}{6}x + 1 \\ y = -\frac{1}{6}x - 1 \end{cases}$		This system has _____ because there is no ordered pair that is on both lines. There is no point of intersection.

- b. Complete the statements.

System

○ A

○ B

○ C

 has no solution because the lines in this system are parallel. Parallel lines have the same

○ slope,

○ y-intercept,

 so there

○ is no point

○ is one point

○ are infinitely many points

 where the lines will intersect.

8.8.3 Guided Practice: Solutions to Systems of Equations

1. Consider the system of equations $\begin{cases} y = \frac{2}{5}x + \frac{6}{5} \\ y = 1.2 + 0.4x \end{cases}$.
- a. Complete the table to determine if each ordered pair satisfies both equations in the system.

Ordered Pair	$y = \frac{2}{5}x + \frac{6}{5}$	$y = 1.2 + 0.4x$	Are Both Equations True?
$(-1.5, 0.6)$			☐ Yes ☐ No
$(0, 1.2)$			☐ Yes ☐ No
$(2, 2)$			☐ Yes ☐ No

- b. Based on your work in Part A, complete the statement.

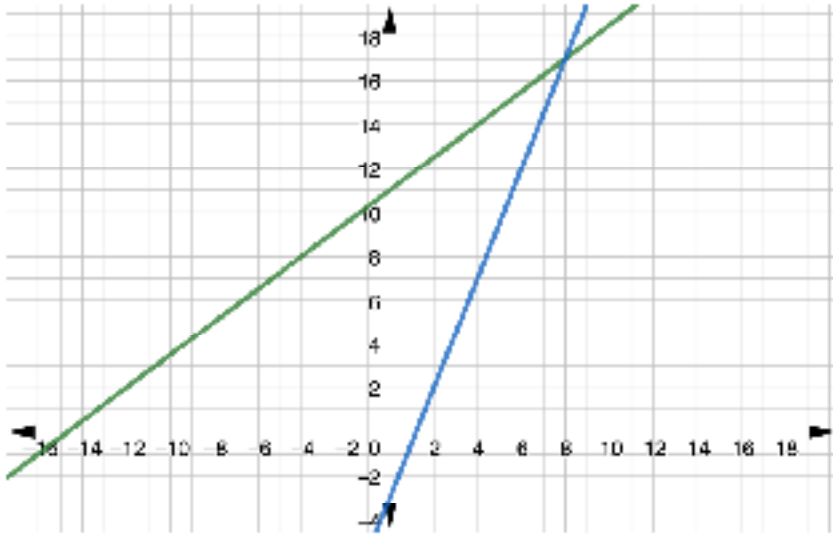
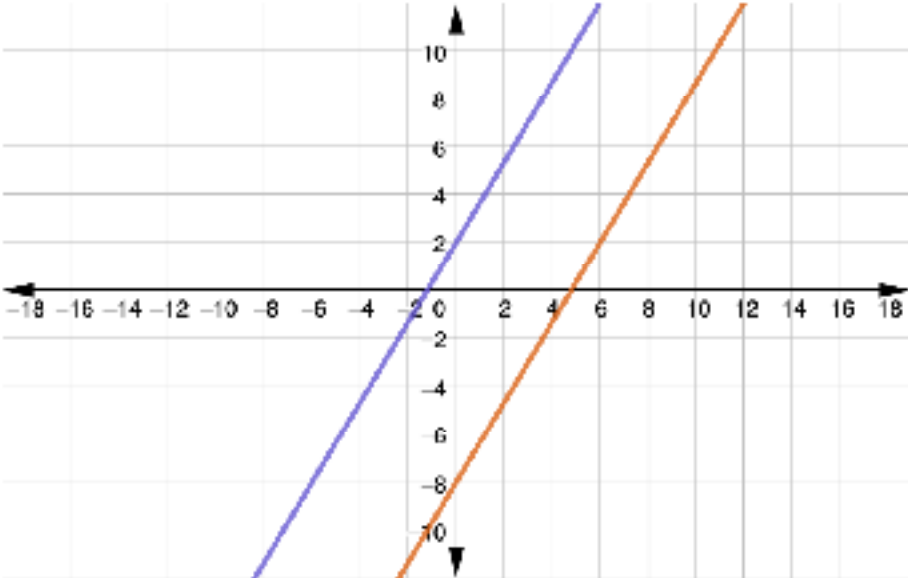
The system $\begin{cases} y = \frac{2}{5}x + \frac{6}{5} \\ y = 1.2 + 0.4x \end{cases}$ has

☐ one solution, _____.

☐ no solutions.

☐ infinitely many solutions.

2. For each system of equations, determine whether there is one solution, no solution, or infinitely many solutions. If there is only one solution, give the solution.

System of Equations	Solution(s)
$\begin{cases} y = \frac{3}{4}x + 11 \\ y = 2.5x - 3 \end{cases}$ 	
$\begin{cases} y = \frac{5}{2}x - 8 \\ y = \frac{15}{9}x + 2 \end{cases}$ 	

8.8.4 Your Turn!

1. Consider the system of equations $\begin{cases} y = \frac{5}{6}x - 2 \\ y = 6 - \frac{1}{2}x \end{cases}$
- a. Complete the table to determine if each ordered pair satisfies both equations in the system.

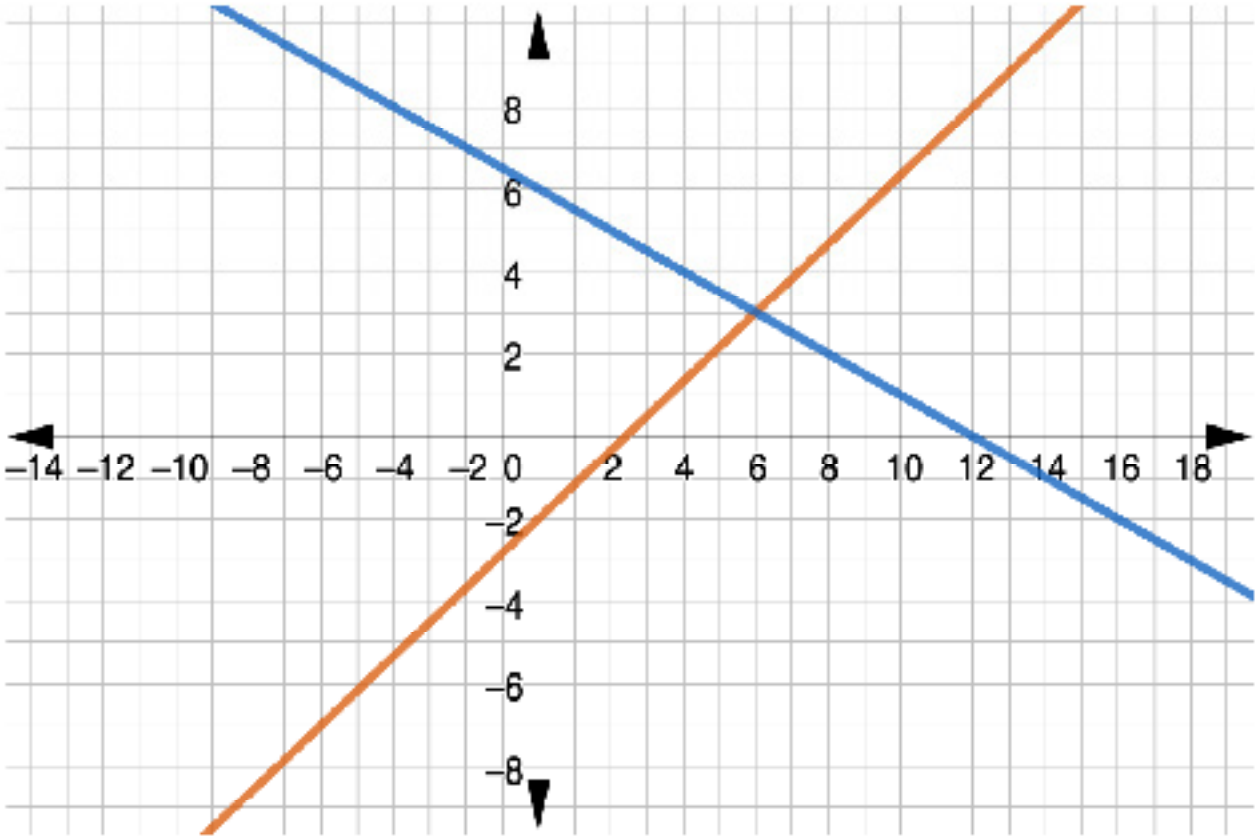
Ordered Pair	$y = \frac{5}{6}x - 2$	$y = 6 - \frac{1}{2}x$	Are Both Equations True?
(4,4)			☐ Yes ☐ No
(6,3)			☐ Yes ☐ No
(12,0)			☐ Yes ☐ No

- b. Based on your work in Part A, complete the statement.

The system $\begin{cases} y = \frac{5}{6}x - 2 \\ y = 6 - \frac{1}{2}x \end{cases}$ has

☐ one solution, _____.
☐ no solutions.
☐ infinitely many solutions.

- c. The graph of the system $\begin{cases} y = \frac{5}{6}x - 2 \\ y = 6 - \frac{1}{2}x \end{cases}$ is shown.



Use the graph to explain why your answer to Part B is correct.

8.8.5 Wrap-Up: Error Analysis

Bevan solved the system of equations $\begin{cases} y = \frac{3}{4}x - 1 \\ y = \frac{1}{4}x - 3 \end{cases}$.

She used the ordered pairs shown in the table to determine the solution(s) to the system. She put a green checkmark in each of the boxes where the ordered pair satisfied the equation.

Ordered Pair	$y = \frac{3}{4}x - 1$	$y = \frac{1}{4}x - 3$
(2, 0.5)	$0.5 = \frac{3}{4}(2) - 1 = \frac{3}{2} - 1$ $0.5 = \frac{1}{2}$ 	$0.5 = \frac{1}{4}(2) - 3 = \frac{1}{2} - 3$ $0.5 \neq -\frac{7}{2}$
(0, -3)	$-3 = \frac{3}{4}(0) - 1 = 0 - 1$ $-3 \neq -1$	$-3 = \frac{1}{4}(0) - 3 = 0 - 3$ $-3 = -3$ 
(-4, -4)	$-4 = \frac{3}{4}(-4) - 1 = -3 - 1$ $-4 = -4$ 	$-4 = \frac{1}{4}(-4) - 3 = -1 - 3$ $-4 = -4$ 

She concluded there are infinitely many solutions to the system because more than one point satisfies the equations.

1. Write a short note to Bevan explaining the error she made.